

Tapiwa Pawandiwa

Full-Stack Software Engineer · Mobile & Web · AI-Powered Products

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SUMMARY

Full-stack software engineer with an MSc in Computer Science and a strong foundation in object-oriented programming, agile software development, and mobile product engineering. I design, build, and ship end-to-end digital products spanning React Native mobile apps, RESTful APIs, and AI-powered backends, with hands-on experience in test-driven development, cross-functional collaboration, and applied machine learning research. I thrive in environments that move fast, value engineering quality, and build products that genuinely matter to users.

EDUCATION

MSc Computer Science · [IU University](#) · [Berlin](#) · [December 2025](#)

1st Class Thesis (Distinction): Benchmarking Explainable AI Methods for Obstetric Clinical Decision Support (BERNA Framework)

Honours, Information Systems

2019 – 2020

[Rhodes University](#) · [South Africa](#)

BSc Computer Science

2017 – 2019

[Rhodes University](#) · [South Africa](#)

EXPERIENCE

Founding Engineer

Jan 2026 – Present

[FoodStorii](#) · [Berlin](#) · Early-stage AI food management startup, joined immediately after MSc graduation

- Architected a full-stack AI-powered mobile application using React Native/Expo, FastAPI, LangGraph, Supabase (PostgreSQL), and Redis, applying object-oriented design principles throughout.
- Integrated the Anthropic Claude API to power an in-app conversational AI assistant for personalised cooking guidance and shopping-list management.
- Implemented production-grade security including rate limiting, row-level security (RLS), Redis key-injection hardening, and PII handling via LangSmith.
- Applied test-driven development and wrote unit and integration tests using Jest. Designed the CI/CD pipeline and managed EAS build distribution for TestFlight (iOS) and Android beta.
- Monitored application performance and error tracking using Sentry and PostHog.

Software Engineer (Working Student)

2023 – 2024

[Volkswagen AG](#) · [Berlin](#)

- Collaborated within a cross-functional agile engineering team delivering internal web applications and RESTful APIs using Svelte, TypeScript, and Go.
- Practised pair programming as a regular team discipline, improving code quality, knowledge sharing, and collaborative problem-solving.
- Applied test-driven development (TDD), writing unit and integration tests to ensure software reliability and maintainability.
- Built and monitored cloud infrastructure using AWS CDK and AWS CloudWatch. Contributed to cybersecurity threat modelling applying secure software design principles.
- Operated in an Agile and Extreme Programming (XP) environment across sprint planning, code reviews, and iterative delivery.

Software Engineering Intern

2019

[Singular Systems](#) · [Cape Town](#)

- Designed and implemented a full-stack job-recruitment portal using object-oriented programming in C#, Microsoft SQL Server, and React JS, delivering both frontend and backend components.
- *Additional:* Microsoft 365 administration and SharePoint configuration for Pawacons water engineering consultancy.

PROJECTS

BERNA Framework

[Clinical AI](#) · [Explainable AI](#) · [ML Robustness Research](#) · [Thesis awarded Distinction](#)

Designed and implemented a systematic benchmarking framework evaluating the stability and robustness of post-hoc XAI methods (SHAP, counterfactuals, prototypes, uncertainty estimation) applied to ML models for fetal-health classification, validated with 17 clinical practitioners. Applied statistical significance testing across perturbation conditions — key finding: XAI method selection is a stronger determinant of robustness than model class.

[Stack](#) Python · PyTorch · Scikit-learn · SHAP | Dataset: UCI CTG

Diner

[Full-Stack Mobile Marketplace](#) · [React Native](#) · [Beta](#)

Built a marketplace mobile app connecting travellers with local hosts offering home-cooked meals, featuring location-based discovery, multi-basket booking, authentication flows, favourites, and device-calendar integration. Managed complex global state across four React Context providers and wrote end-to-end and unit tests using Jest.

[Stack](#) React Native · Expo · AWS Amplify (DynamoDB, GraphQL, S3, Cognito) · TypeScript · React Query · Node.js

ShareWhere

[Full-Stack Mobile Donations Platform](#) · [React Native](#) · [Beta](#)

Built a hyper-local item-based donations platform with a 24-hour commitment timer, replacing a cron-based polling architecture with a fully event-driven system using Supabase database triggers and Edge Functions. Implemented dual-portal access, row-level security, and real-time status synchronisation.

[Stack](#) React Native · Expo · Supabase (PostgreSQL, RLS, Edge Functions, Triggers) · React Query · TypeScript

SKILLS

MOBILE & FRONTEND React Native, Expo, React, Next.js, TypeScript, JavaScript, Tailwind CSS, Svelte, HTML/CSS

UI/UX & DESIGN Figma, UI/UX Design, Prototyping, Design Systems

BACKEND & DATA Python, FastAPI, C#, Go, PostgreSQL, Redis, REST APIs, GraphQL, SQL, Node.js

AI & MACHINE LEARNING PyTorch, Scikit-learn, LangGraph, Anthropic Claude API, Explainable AI, SHAP, Counterfactuals, Uncertainty Estimation, Statistical Significance Testing, ML Robustness Evaluation, LangSmith

TESTING & QUALITY Jest, Unit Testing, Integration Testing, End-to-End Testing, Test-Driven Development (TDD)

INFRASTRUCTURE & TOOLS AWS CDK, AWS Amplify, AWS CloudWatch, Supabase, Sentry, PostHog, Microsoft 365, SharePoint, Docker, Render, EAS Build, GitHub, GitHub Actions

METHODOLOGIES Agile, Extreme Programming (XP), Pair Programming, Object-Oriented Programming (OOP), CI/CD, Secure Software Design

CERTIFICATIONS

- Machine Learning, Intermediate · Kaggle
- Data Science with Python · University of Cape Town

LANGUAGES

English — Native · German — A2, actively improving